

Qatar-4b

R-2481

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_211219 · created 2026-08-24T00:13:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459567.7158 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.044
Transit depth (Rp/Rs)^2	3.39 +/- 1.6 [%]
Orbital Inclination (inc)	89.89 +/- 1.98
Airmass coefficient 1 (a1)	0.873 +/- 0.06
Airmass coefficient 2 (a2)	0.092 +/- 0.034
Scatter in the residuals of the lightcurve fit is	7.04 %
Best Comparison Star	#1 - [428, 39]
Optimal Method	PSF photometry
Transit Duration (day)	0.094 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.184 ± 0.044 vs 0.138 (deviation 0.64σ)
Duration fit vs published	0.094 d vs 0.0892 d (deviation 0.62σ)
Mid-time O–C	9.4 min
Depth SNR	2.6
Residual scatter	7.04 %

Light curve

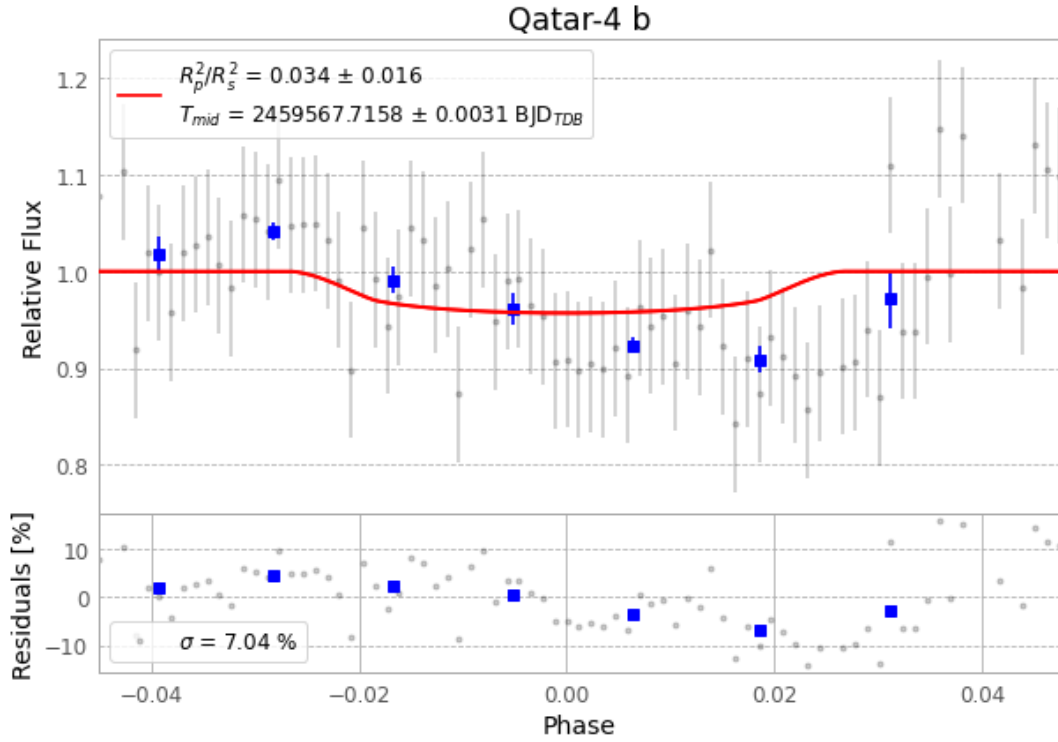


Figure 1 — FinalLightCurve_Qatar-4 b_2021-12-18.png (SHA-256 49957bf443538c25...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [335, 271], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5c232700759e9cb5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

80 FITS frames (53 MB), Qatar-4211219030911.FITS → Qatar-4211219070912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-211219.log (1ed34f61bd4a...), FinalLightCurve_Qatar-4 b_2021-12-18.png (49957bf44353...), FinalLightCurve_Qatar-4 b_2021-12-18.csv (91c4d55988a4...)

Compute resources

Wall time 102.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 00:13Z.